Trainings

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

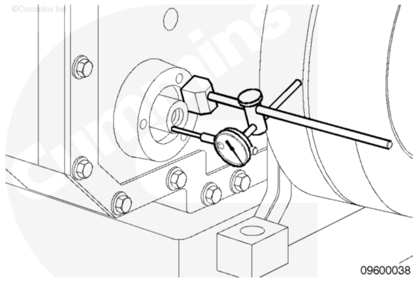
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the water pump. Refer to Procedure 008-062 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-062-tr.html)
- Remove the alternator drive pulley. Refer to Procedure 009-010 in Section 9. (/qs3/pubsys2/xml/en/procedures/56/56-009-010-tr.html)
- Remove the alternator drive seal. Refer to Procedure 001-001 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-001.html)

Initial Check

with Mechanically Actuated Injector

Measure the water pump drive end clearance after the drive has been installed.

Water Pump Drive End Clearance - Installed		
mm		in
0.05	MIN	0.0020
0.21	MAX	0.0083



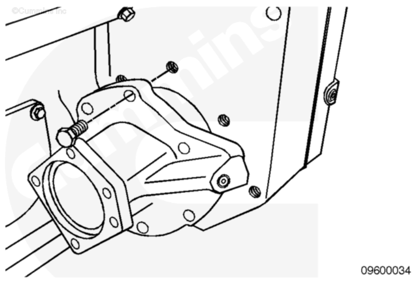
Remove

with Mechanically Actuated Injector

Remove the six capscrews.

Lightly tap on the end of the shaft with a rubber mallet, and remove the water pump drive assembly.

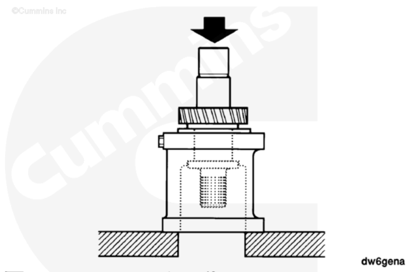
Remove, inspect, and discard the gasket, if damaged.



Disassemble

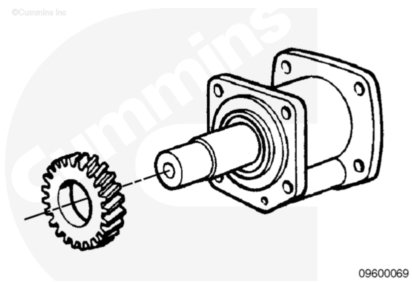
with Mechanically Actuated Injector

Support the housing and press the shaft through the gear until the gear is free from the shaft.



Remove the pipe plug from the housing.

Use a light to inspect the main oil passage. The passage **must** be clean and open.



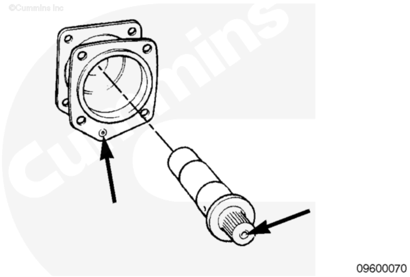
Remove the shaft.

Check the housing for an orifice expansion plug.

Check the shaft for an orifice pipe plug.

The plugs **must** be removed to clean the oil drillings if dirt or debris is found in the main oil passage.

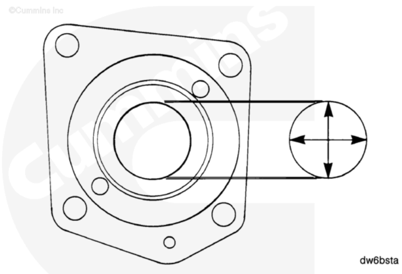
Do **not** remove the expansion plugs if the main oil passage is clean.



Measure the inside diameter of the housing bushings.

Water Pump Drive Bushing Diameter			
	mm		in
Flanged Bushings	49.95	MIN	1.965
	50.03	MAX	1.9697
Standard Bushings	44.48	MIN	1.7512
	44.58	MAX	1.7551

If the bushings are **not** within specifications, the bushings **must** be replaced.

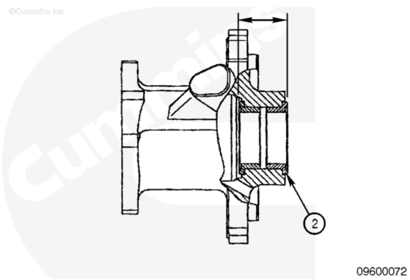


Check the grooved surface of the thrust faces for damage.

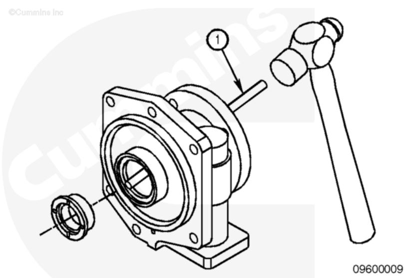
Measure the distance from the thrust bearing faces (2).

Water Pump Drive Thrust Bearing Face Distance		
mm		in
45.370	MIN	1.7862
45.450	MAX	1.7894

If the thrust bearing face distance is **not** within specifications, the bushings **must** be replaced.



Remove the flanged bushing from the drive housing using a brass drift (1), taking care **not** to damage the bushing.

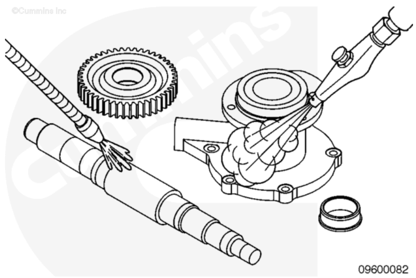


Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

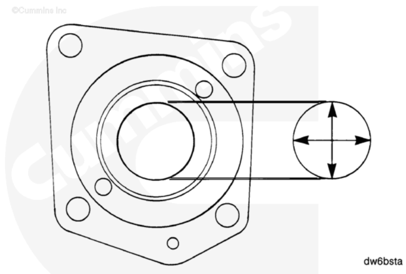
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the parts with solvent.

Measure the inside diameter of the housing bushing bore.

Water Pump Drive Housing Bushing Bore		
mm		in
60.000	MIN	2.3622
60.050	MAX	2.3642

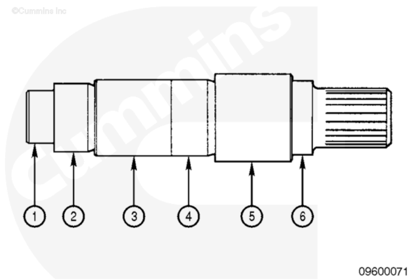


If the housing bushing bore is **not** within specifications, the housing **must** be replaced.

Check the shaft at the seal area for wear.

Check the shaft splined area for wear or damage.

Measure the outside diameter of the shaft.

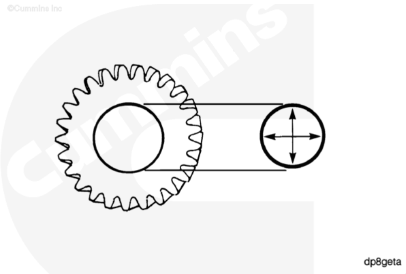


Water Pump Drive Shaft Outside Diameter			
	mm		in
(1)	35.084	MIN	1.3813
	35.096	MAX	1.3817
(2)	43.928	MIN	1.7294
	43.940	MAX	1.299
(3)	44.370	MIN	1.7468
	44.630	MAX	1.7571
(4)	45.070	MIN	1.7744
	45.082	MAX	1.7749

(5)	49.820	MIN	1.9614
	49.832	MAX	1.9619
(6)	39.994	MIN	1.5746
	40.006	MAX	1.5750

If the shaft is **not** within specifications, the shaft **must** be replaced.

Check the gear teeth for wear or pitting.



Measure the inside diameter of the gear.

Water Pump Drive Gear Inside Diameter		
mm		in
44.98	MIN	1.7709
45.02	MAX	1.7724

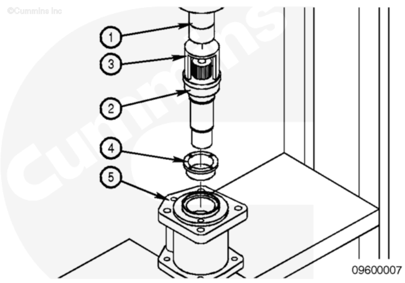
If the gear is **not** within specifications, the gear **must** be replaced.

Assemble

with Mechanically Actuated Injector

⚠ CAUTION ⚠

The bushings must be pressed in square to the bores of the housing. The thrust faces of the two bushings must be parallel and concentric. Failure to comply can result in distortion of the thrust face.

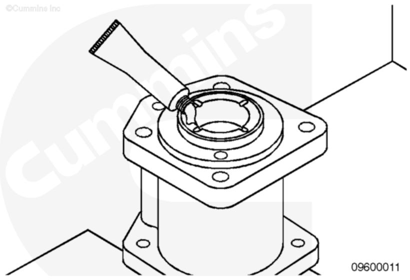


Support the housing (5).

Using an arbor press (1), spacer (3), and shaft (2), press the bushing (4) into the housing.

Repeat the process for the remaining bushing.

Lubricate the bushing bores and thrust faces with Lubriplate™ 105, or equivalent.

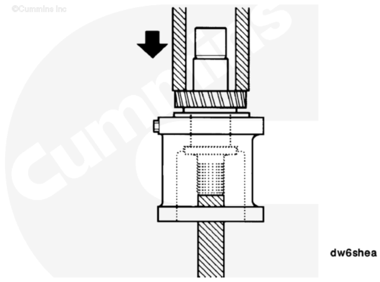


Support the housing and shaft assembly in a hydraulic press.

Clean any Lubriplate™ from the shaft and apply an even coat of Loctite® 609, or equivalent, to the inside diameter of the gear.

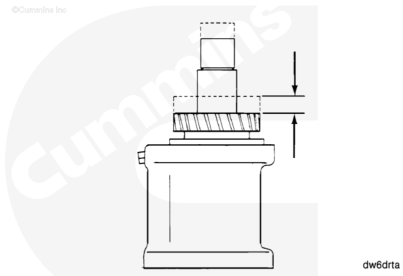
Install the shaft into the housing.

Press the gear onto the shaft until the gear touches the shoulder of the shaft.



Measure the water pump drive end clearance.

Water Pump Drive End Clearance		
mm		in
0.05	MIN	0.0020
0.21	MAX	0.0083



If the end clearance is **not** within specifications, make sure that both thrust bearings are correctly positioned in the water pump drive housing counterbores.

If the bearings are correctly positioned and the end clearance is **not** within specifications, the drive **must** be completely disassembled and individual component dimensions checked.

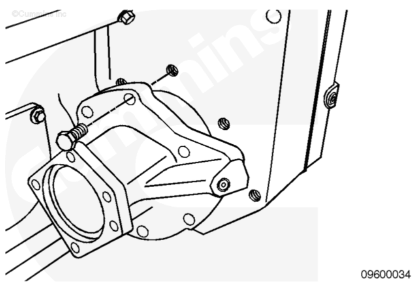
Install

with Mechanically Actuated Injector

Lubricate the bushing in the front gear housing cover with Lubriplate™ 105, Part Number 3163086, 3163087, or equivalent.

Install the gasket on the drive pilot.

Install the pump drive, allowing the gear to turn as it meshes with the idler gear.



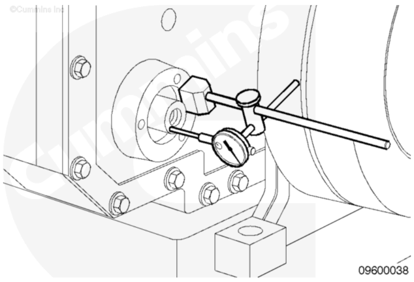
Align the capscrew holes, and install the six water pump drive mounting capscrews.

Tighten the six capscrews.

Torque Value: 45 n•m [33 ft-lb]

Measure the water pump drive end clearance after the drive has been installed.

Water Pump Drive End Clearance - Installed		
mm		in
0.05	MIN	0.0020
0.21	MAX	0.0083



Finishing Steps

with Mechanically Actuated Injector

- Install the alternator drive seal. Refer to Procedure 001-001 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-001.html)
- Install the alternator drive pulley. Refer to Procedure 009-010 in Section 9 (/qs3/pubsys2/xml/en/procedures/56/56-009-010-tr.html).
- Install the water pump. Refer to Procedure 008-062 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-062-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Operate the engine and check for leaks.